

Mathematical Operations

A common requirement is to perform mathematical operations, and Python is actually quite powerful at this. Python can work with arbitrarily large numbers, so if you were just looking for a calculator to calculate 2^{2000} , Python should have the answer for you (all 604 digits of it) beforekwin you can count to, well one.

All common operation such as addition, subtraction, multiplication and division work as you'd expect using their usual symbols (+, -, *, /). The modulo operator also uses the conventional '%' character. Exponentiation is performed using two asterisks, as: 2^{**2000}

On difference between Python 2 and 3 here is that Python 3 will output $3/2$ as 1.5 while Python 2 will floor that to 1.



```

2**2000
114813469527425453421283320117768196462231770208899520947764273882
578826139237031385685948531850626691844594453896746277344711896388
305533142508135616685318539129889114531228000688739148280346871438
92699006348824478141548364638836394731702604046635397093499656162
3088089448236364233116496361642219763326813441689688646485654321378
48480791405890632477650042960271679462436522008132236281261761263
883317206598995396416127021779858404342156853183251540886433603061
0205549577835850720301660819572166305877553604255483776015528348789
419432054508915275793882025175439526900822842770817966452762184851
1480293176

```

Complex mathematical operations are a cakewalk with Python

String Operations

Web application especially need to be able to handle string manipulations. Here too you will find Python's syntax quite beneficial.

Like with most programming languages, Python strings are surrounded by quotes. If you need to use a single quote in your string, use double quotes to encapsulate it and vice versa. If you need both, you can escape the quote character using a backslash, as you can see in the following examples of Python strings:

- "A simple string"
- 'There are no strings on me'
- "Of course" said noone in particular'
- "Here's a string"
- "Here's a string" said the previous line'

Like with many languages, you can use '\t' for tabs and '\n' for the newline character.

Another convenience Python provides when dealing with long strings, is the triple quote. If you want to enter a para of text with all kinds of quotation marks and new lines, rather than littering it with escape characters you can use the triple quote as follows:

- some_html = """